

29.07.2024

LSM

- ① Inference on Simple Linear Regression
- ② Diagnostic on Simple Linear Regression

Measure of Quality of fit

Residuals $e_i = Y_i - \hat{Y}_i$, $i=1, \dots, n$

$\underline{e} \in \mathbb{R}^n$ has the following properties

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$e_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$$

e_i can be seen as an estimate of ϵ_i

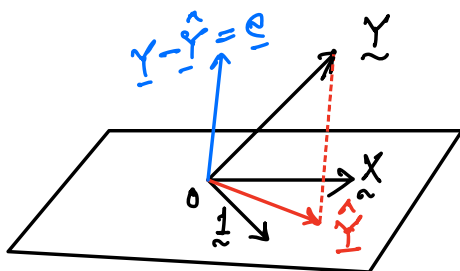
$$\left. \begin{array}{l} (i) \sum e_i = 0 \\ (ii) \sum X_i e_i = 0 \end{array} \right\} \Rightarrow \text{Normal Equation}$$

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$

$\Rightarrow$ (a) e_i 's are not independent
(b) $\underline{1}^T \underline{e} = 0$, $\underline{X}^T \underline{e} = 0$

$\underline{e}$ subspace orthogonal to $\underline{X}_{n \times 2} = [\underline{1} \ \underline{X}]$

degrees of freedom of $\underline{e} = n - 2$
 $= n - \text{Rank}(\underline{X})$



$$\begin{aligned} L(u_0, u_1) &= \sum_i (Y_i - (u_0 + u_1 X_i))^2 \\ &= \|\underline{Y} - (u_0 \underline{1} + u_1 \underline{X})\|_2^2 \end{aligned}$$

$$\underline{Y} = \hat{\underline{Y}} + \underline{e}$$

↑ ↑
mutually orthogonal

$$\hat{Y} = \begin{pmatrix} \hat{Y}_1 \\ \vdots \\ \hat{Y}_n \end{pmatrix} \leftarrow \text{vector of fitted values}$$

$$\hat{Y} = \hat{\beta}_0 \underline{1} + \hat{\beta}_1 X = [1: X] \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = X \hat{\beta}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \underbrace{X(X^T X)^{-1} X^T}_{P_X} Y$$

P_X satisfies

$$(i) P_X^T = P_X$$

$$(ii) P_X^2 = P_X \text{ (idempotent)}$$

$$(iii) \mathcal{L}(P_X) = \mathcal{L}(X)$$

P_X is the orthogonal projection matrix (projector) onto $\mathcal{L}(X)$

We also call P_X the "hat matrix" associated with the regression problem

$$\begin{aligned} Y &\mapsto P_X Y = \hat{Y} & Y &= P_X Y + (I - P_X) Y \\ &\searrow (I - P_X) Y = e & &= \hat{Y} + e \end{aligned}$$

$$SSE = \sum_{i=1}^n e_i^2 = \|e\|_2^2 = \|(I - P_X) Y\|_2^2$$

$$SSE = Y^T (I - P_X) Y$$

$$df(SSE) = df(e) = n - 2$$

$$\underline{Y} = \hat{\underline{Y}} + \underline{e}$$

$$\Rightarrow \underline{Y} - \bar{Y}\underline{1} = (\hat{\underline{Y}} - \bar{Y}\underline{1}) + \underline{e}$$

Squared norm

$$\begin{aligned} \sum (Y_i - \bar{Y})^2 &= \|\hat{\underline{Y}} - \bar{Y}\underline{1}\|_2^2 + \|\underline{e}\|_2^2 = \sum (\hat{Y}_i - \bar{Y})^2 + SSE \\ &\quad \downarrow \\ &= \sum (\hat{Y}_i - \bar{Y})^2 = \sum (\hat{\beta}_0 + \hat{\beta}_1 X_i - \hat{\beta}_0 - \hat{\beta}_1 \bar{X})^2 + SSE \\ &= \sum (\hat{Y}_i - \hat{\bar{Y}})^2 = \hat{\beta}_1^2 \sum (X_i - \bar{X})^2 + SSE \\ &\quad \uparrow_{\text{since } \underline{e}_i = 0} \end{aligned}$$

ANOVA decomposition of $\underline{Y}$

$$\begin{aligned} \sum (Y_i - \bar{Y})^2 &= \sum (\hat{Y}_i - \bar{Y})^2 + \sum (Y_i - \hat{Y}_i)^2 \\ \text{SSTo} &= \hat{\beta}_1^2 \sum (X_i - \bar{X})^2 + SSE \\ &= \overbrace{\quad}^{\text{SSR}} + SSE \end{aligned}$$

Relative contribution of X in describing variability in Y (through the linear regression model) is expressed

as
$$R^2 = \frac{SSR}{SSTo} = 1 - \frac{SSE}{SSTo}$$

multiple correlation coefficient

Note
$$R^2 = \frac{\hat{\beta}_1^2 \sum_i (X_i - \bar{X})^2}{\sum_i (Y_i - \bar{Y})^2} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}} = r_{XY}^2$$

Inference in linear regression model :

Under the assumption $E\varepsilon_i = 0$ for all i :

We checked that $E\hat{\beta}_0 = \beta_0$, $E\hat{\beta}_1 = \beta_1$

To answer question about $\begin{cases} \text{interval estimating} \\ (\beta_0, \beta_1) \text{ or some function of} \\ \text{hypothesis testing about} \\ \text{prediction of } \bar{Y} \text{ for a given} \\ \text{value of } \bar{X} \end{cases}$

To be able to answer this inferential question, we need to describe the probability distribution of the following statistics $(\hat{\beta}_0, \hat{\beta}_1, SSE) \equiv (\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2)$ where $\hat{\sigma}^2 = MSE = \frac{SSE}{n}$

To ensure that we have a description of the sampling distribution of $(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2)$

If we assume that $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$, then we have a complete description of the sampling distribution

Result: $\underset{\text{under}}{\hat{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \sim N(\underline{\beta}, \sigma^2 (X^T X)^{-1})$

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-2}$$

and $\hat{\beta}$ and SSE are independently distributed

$$SSE = Y^T (I - P_X) Y = \underbrace{\varepsilon^T (I - P_X) \varepsilon}_{\text{ev. } (n-2) \text{ 1's, 2 0's}}$$

By Spectral Decomposition

$$P_X = Q \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix} Q^T$$

$$\frac{1}{\sigma^2} e^T (I - P_X) e = \tilde{z}^T Q \begin{pmatrix} 0 & 0 \\ 0 & I_{n-2} \end{pmatrix} Q^T \tilde{z}$$

$$= \sum_{i=3}^n \tilde{z}_i^2 \sim \chi_{n-2}^2$$

$$\tilde{z} \sim N(0, I_n)$$

$$SSE = \|(I - P_X)Y\|^2$$

$$\tilde{z} = Q^T z \sim N(0, I_n)$$

$\hat{\beta} = (X^T X)^{-1} X^T Y$. Since $X^T Y$ and $(I - P_X)Y$ are jointly normal and uncorrelated $\Rightarrow$ they are independent.

Function of $X^T Y$

So $\hat{\beta}$ and e are independent

Inference Questions

Confidence Interval for β_0 and β_1

$$\frac{\hat{\beta}_j - \beta_j}{S(\hat{\beta}_j)}$$

$S^2(\hat{\beta}_j)$ estimate of variance of $\hat{\beta}_j$

$$= \frac{MSE}{\sigma^2} \text{Var} \hat{\beta}_j$$

$$= \frac{\hat{\beta}_j - \beta_j / \sqrt{\text{Var} \hat{\beta}_j}}{\sqrt{\frac{MSE}{\sigma^2}}}$$

$$\xrightarrow[N(0,1)]{\sqrt{\chi_{n-2}^2/n-2}} \text{ indep } \Rightarrow t_{n-2}$$

Estimating the mean of Y given $X = X_h$ (specific value)

$$\mu_h = \beta_0 + \beta_1 X_h$$

Point estimate $\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h = \begin{pmatrix} 1 \\ X_h \end{pmatrix}^T \hat{\beta}$

$$\hat{\mu}_h \sim N\left(\underbrace{\beta_0 + \beta_1 X_h}_{\mu_h}, \sigma^2 \begin{pmatrix} 1 \\ X_h \end{pmatrix}^T (X^T X)^{-1} \begin{pmatrix} 1 \\ X_h \end{pmatrix}\right)$$

Observe that

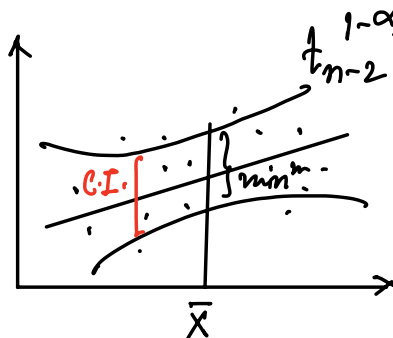
$$\hat{\mu}_h = \hat{\beta}_0 + \hat{\beta}_1 X_h = \bar{Y} + \hat{\beta}_1 (X_h - \bar{X})$$

$$\text{Var} \hat{\mu}_h = \text{Var} \bar{Y} + (X_h - \bar{X})^2 \text{Var} \hat{\beta}_1$$

$$\therefore \hat{\mu}_h \sim N\left(\underbrace{\beta_0 + \beta_1 X_h}_{\mu_h}, \sigma^2 \left(\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2} \right)\right)$$

100(1- α)% CI

$$\hat{\mu}_h \pm \underbrace{t_{n-2}^{-1} \left(1 - \frac{\alpha}{2}\right)}_{1-\alpha/2} \sqrt{MSE} \sqrt{\frac{1}{n} + \frac{(X_h - \bar{X})^2}{\sum (X_i - \bar{X})^2}}$$



Min^r at $\bar{X}$, increases as away from $\bar{X}$.